Geometric Median

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1 Introduction

1.1 Introduction and rationale

The Fermat point (or Torricelli point) in a triangle is the point which minimises the sum of distances to each vertex and has many applications in real life such as in minimising distances in networks. This project aims to explore a variant of the geometric median in triangles, which is when the product is maximised whilst originally the sum is minimised. Such a point of max product may have special properties that can be used in real life, for example in applications like finding structural weakness.

1.2 Research Questions

For each research question, the point is to be found and a formula for its value created. Research Question 1 is included as a precursor to our main Research Questions 2, 3, 4.

1. What point in a triangle minimises the sum of distances?
2. What point in an equilateral triangle maximises the product of distances?
3. What point in an isosceles triangle that maximises the product of distances?
4. What point in a general triangle maximises the product of distances?

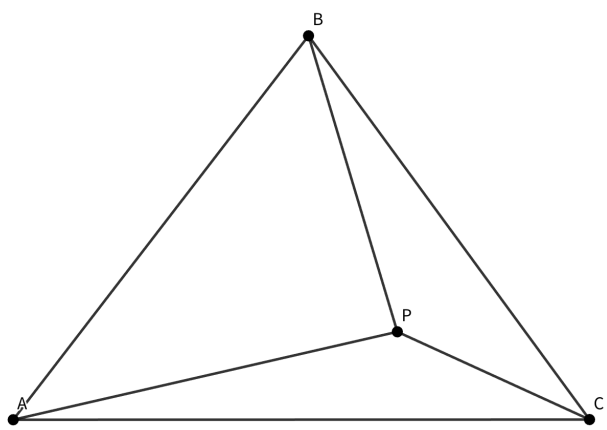
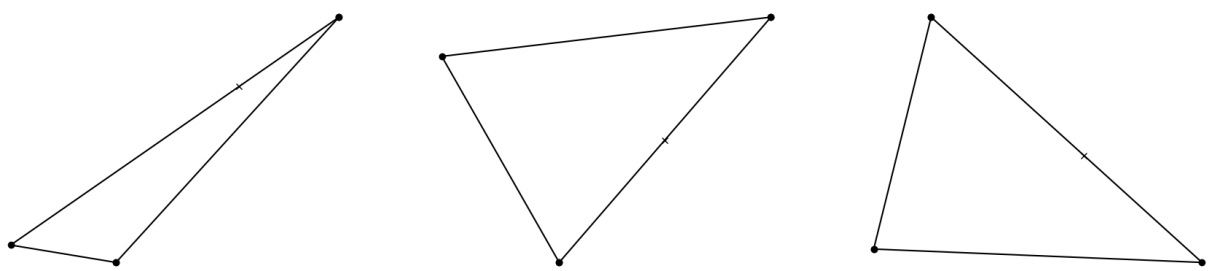


Figure 1: Figure showing a random point P in the interior of a triangle ABC. Research question 1 asks to minimize $|AP| + |BP| + |CP|$. Research questions 2, 3, and 4 ask to maximize $|AP| \cdot |BP| \cdot |CP|$

2 Methodology

In Research Question 1, the Fermat point was solved synthetically using similar triangles and circles.
In Research Question 2, the equilateral triangle case was solved by smoothing to show optimality in shifting the point towards the midpoint of the side.
In Research Questions 3 and 4, the isosceles triangle and general triangle case is too complex to use smoothing. By analysing the various related literature reviews it was established that the point which maximises the product of distances within any n-gon always lies on the boundary of the n-gon. Hence for Research Questions 3 and 4, a generalised formula to locate the point of maximum product in isosceles and general triangles were derived through differentiation and coordinate geometry.
Furthermore, to aid in finding the general location of the maximum point in Research Questions 2, 3 and 4, a bruteforce program was also used to approximate the maximum point. By generating hundreds of random triangles, we observed that the max point should lie on the longest edge of any triangle, which was proven in Research Question 4.



(a) (b) (c)

Figure 2: Random triangles generated by bruteforce program. Max product point is denoted by 'x'

3 Literature Review

While the Fermat point has been thoroughly explored, which refers to finding the minimum sum of distances from the vertices of a triangle to a certain point within the triangle, little research and further studies were conducted to investigate the possibility of finding a point within the triangle that maximises the product from the vertices of a triangle to itself. In this project, we aim to explore this, and shed some light on how to find such a point.
A key theorem we used was that the maximum point will always lie on the edge of any polygon, shown by Binyamin Schwarz in 1965. Furthermore, Gulbank D. Chakerian and Murray S. Klamkin, 2007, also explored the regular n-gon.

4 Results and Findings

4.1 Research Question 1: Minimising Sum of Distances in General Triangle

The point which minimises the sum of distances to the 3 vertices of a triangle is known as the Fermat point. We will show one way to construct the Fermat point and provide a formula for the minimum sum of distances in a triangle.

4.1.1 Construction of Fermat point:

Discuss 2 cases, firstly when there exists an angle $\geq 120°$ and secondly, when there is not.

Case 1: There exists an angle $\geq 120°$ in $\triangle ABC$ **Claim:** The obtuse vertex is the Fermat point, by showing that deviation of the point from the obtuse vertex results in an increase in sum of distances to the vertices.
Detailed working found in Appendix 7.1.1.1

Case 2: All angles are $< 120°$ in $\triangle ABC$ **Claim:** The Fermat point can be found as follows: Construct three equilateral triangles out of the three sides from $\triangle ABC$, then connect each new vertex to the respective opposite vertex. The added lines will concur at the Fermat point.
Detailed working found in Appendix 7.1.1.2

4.1.2 General equation for Fermat point value:

Let the 3 sides of the triangle $\triangle ABC$ be a, b, c, and without loss of generality $a \geq b \geq c$

Case 1: There exists an angle $\geq 120°$ in $\triangle ABC$ Since point F is at vertex A, Fermat point value $= b + c$

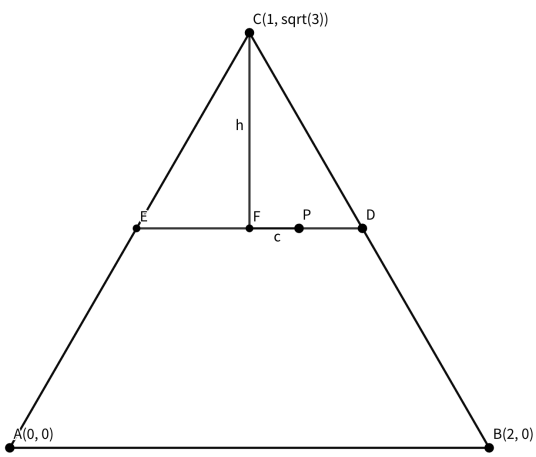
Case 2: All angles are $< 120°$ in $\triangle ABC$ We use the cosine rule then Heron's formula to substitute for area of triangle ABC in terms of a, b, c, we find Fermat point value:

$$\frac{1}{2}\sqrt{a^2 + b^2 + c^2 + \sqrt{3(a+b+c)(-a+b+c)(a-b+c)(a+b-c)}}$$

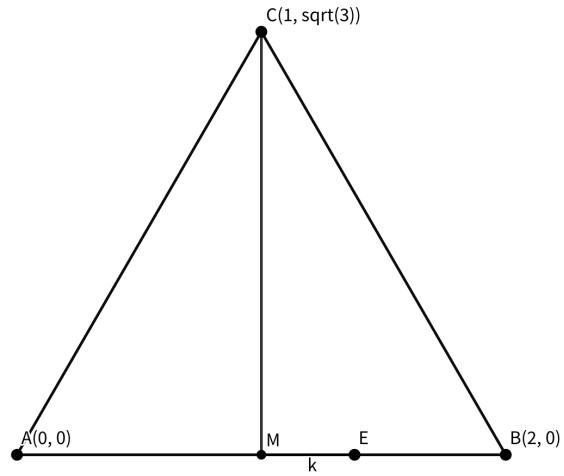
Detailed working found in Appendix 7.1.2

4.2 Research Question 2: Maximising Product of Distances in Equilateral Triangle

The following section consists of 2 parts.



(a) In 4.2.1, it is proven that the max point does not lie in the interior of the triangle.



(b) In 4.2.2, it is proven that the max point lies at the midpoint on the edge.

Figure 3

4.2.1 Max point lies on the boundary of the triangle

Claim: Point on boundary will yield greater product of distances than point within boundary
Steps taken:

1. Consider a variable point P in the interior of the triangle.
2. Construct a line ED parallel to a side of the triangle through P.
3. Set up product expression of point P, in terms of distance from midpoint F of ED.
4. Differentiate product expression and do second derivative test.
5. It is obtained that midpoint F of line ED is a minimum point.
6. Hence maximum value will be attained when point P is shifted along ED onto the boundary of the triangle.

Proof found in Appendix 7.2.1

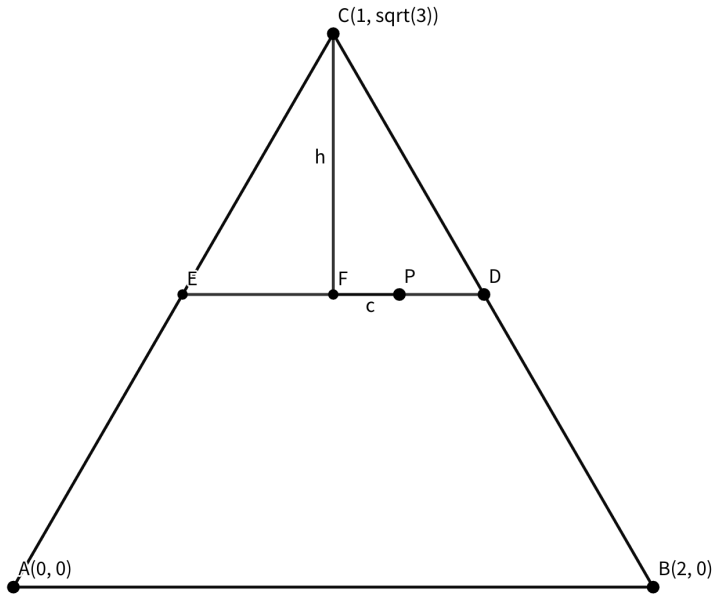


Figure 4: Product value for points along line ED is minimum at point F. Hence, the maximum occurs on the boundary at point D

4.2.2 Max point lies on the midpoint of the edge

Claim: Max point lies on the midpoint of the edge
Steps taken:

1. Set up product expression of point E, in terms of distance from midpoint M of a side.
2. Smooth the product expression towards the midpoint.
3. It is obtained that the product expression increases towards midpoint, hence, the point of max product in equilateral triangle is obtained at the midpoint of the sides.

Proof found in Appendix 7.2.2

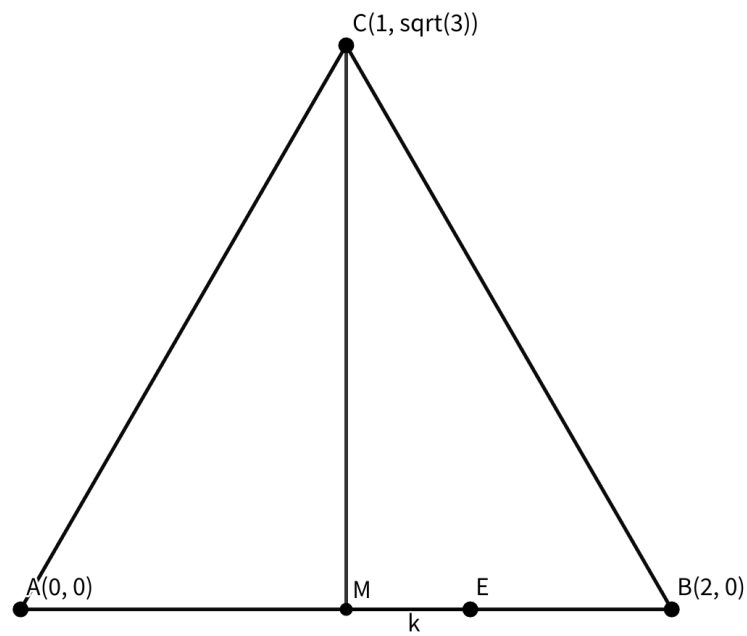


Figure 5: At point $E \neq M$, the product value at E is less than product value at M, i.e. product value is greatest at midpoint M

Hence, the formula for max product in equilateral triangle with side length $a = \frac{\sqrt{3}a^3}{8}$

4.3 Research Question 3: Maximising Product of Distances in Isosceles Triangle

According to Binyamin Schwarz in 1936 [**?**] and Gulbank D. Chakerian with Murray S. Klamkin in 2007 [**?**], the point which maximises the product of distances within any n-gon always lies on the boundary of the n-gon.
The max value on the base and leg is calculated separately, then compared to obtain the max value in the entire triangle.

4.3.1 Calculating max point on the base

Claim:

$$\text{Max product value on base} = \begin{cases} t, & \text{if } t > \frac{1}{\sqrt{2}} \\ \dfrac{2\sqrt{(1-2t^2)^3}}{3\sqrt{3}}, & \text{otherwise} \end{cases}$$

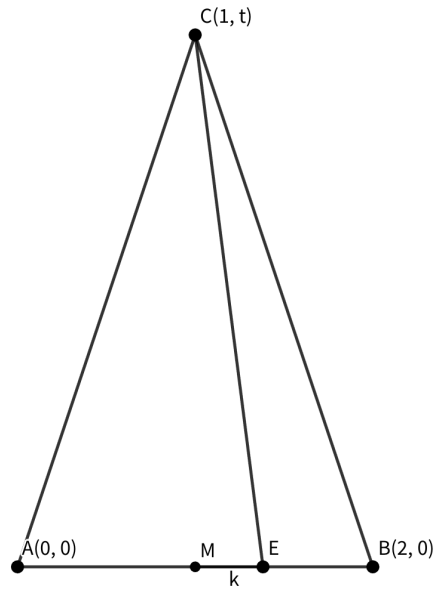


Figure 6: Isosceles triangle. Max value on the base depends on height t.

Steps taken:

1. Set up expression of point E, in terms of distance from midpoint M of side AB
2. Differentiate product expression and do second derivative test

Proof found at Appendix 7.3.1

4.3.2 Calculating max point on the leg of isosceles

Claim: There is only one maximum point on the leg, with exact value (1).
Proof found at Appendix 7.3.2

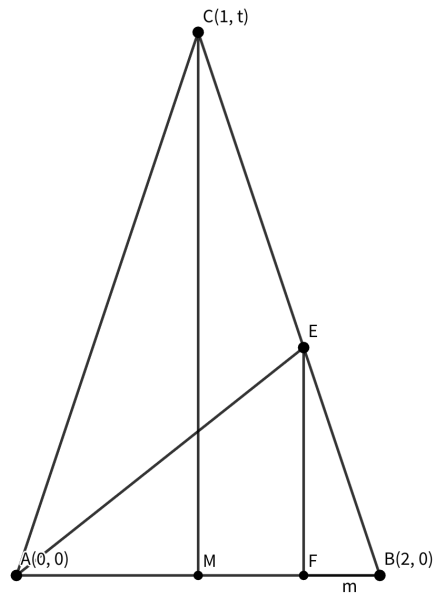


Figure 7: Max value on leg can be derived using differentiation.

4.3.3 Comparing max on leg and max on base

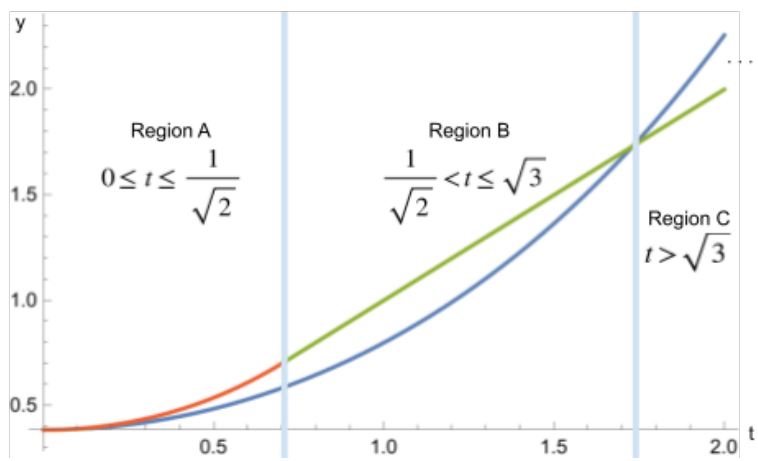


Figure 8: Red line and Green line represent the maximum product of distances to vertices when point E is on base. Blue line represents max product of distances to vertices when point F is on side.

Note that the blue line intersects the green line at $t = \sqrt{3}$. This means that when the value of $t > \sqrt{3}$ (Region C), maximum product of distances on the side is always greater than that of the base. Also, when $t < \sqrt{3}$ (Region A and B), the maximum product of distances on the base is always greater than that of the side as shown by the graph. Note that when $t = \sqrt{3}$, side length of the isosceles triangle is equal to the base length of the isosceles triangle.
Hence, it is shown that the point which maximises the product of the distances to the vertices is found on the longest side of the isosceles triangle.

4.4 Research Question 4: Maximising Product of Distances in General Triangle

4.4.1 Calculating max points on each side

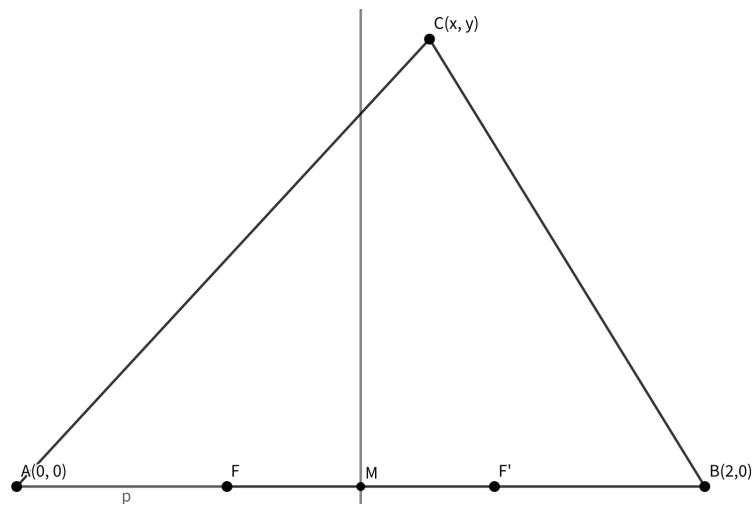


Figure 9: Without loss of generality $|AB| \geq |CA| \geq |BC|$. Location of the max point can be bounded to be in the region to the right of the perpendicular bisector of side AB.

We calculate the max values on each side as follows:

1. Bound the location of the max point by reflection across midpoint
2. Differentiate for all three sides
3. Solve derivative $= 0$ for all three sides

Proof found at Appendix 7.4.1

4.4.2 Comparing maximums on each side of general triangle

3D graph is used to compare the maximum values of the product of distances when the point lies on AB, BC and CA.

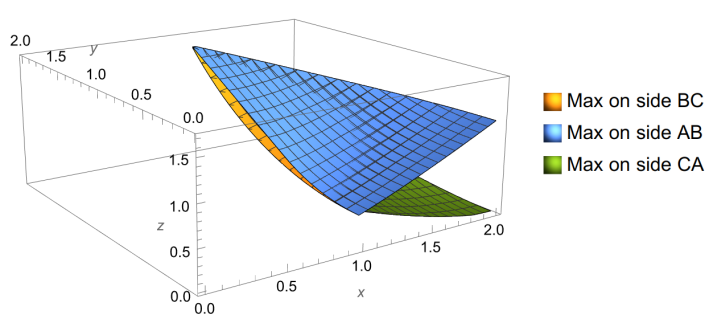


Figure 10: It can be observed that the max value on side AB is always greatest

The equations, as well as the analysis of the graph, can be found in Appendix 7.4.2.
It can be observed that the blue plane which represents the max on side AB, the longest side, is always on top, i.e. max point lies on the longest side.
Hence, the solution for p on the longest side, which we have found previously, is the max point in a general triangle. The expression of p was derived in Appendix 7.4.1.

5 Conclusion

5.1 Summary

The point at which the sum of distances between all vertices and said point is minimised in a triangle is known as the Fermat point, and can be constructed using synthetic methods.
The point at which the product of distances between all vertices and said point is maximum always lies on the longest side of the triangle. It is possible to locate this point in any triangle.

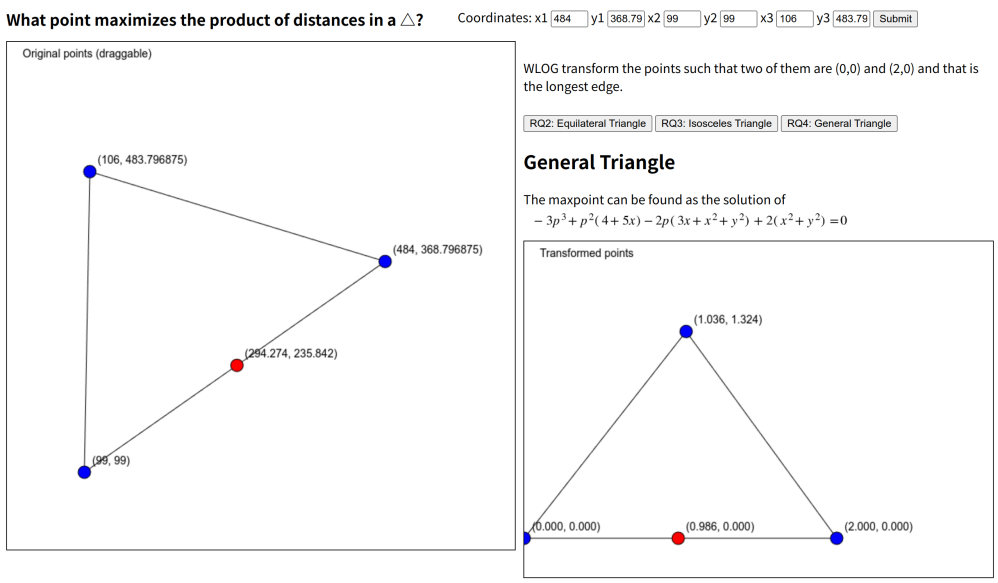


Figure 11: Our results are demonstrated in a program that calculates the max point in a general triangle. Download the html file here: . Points can be input in the left canvas. Location of the max point can be found as the solution of the cubic equation.

5.2 Possible extensions

Further investigations to locate the point of maximum product of distances in n-gon as well as formula to find the maximised product could be carried out in the future.

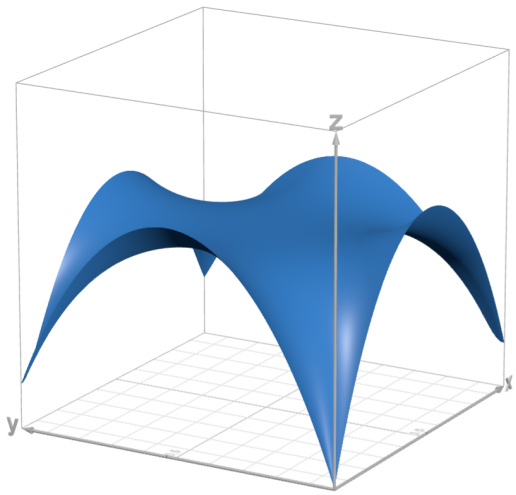


Figure 12: Graph of product value at different points of a square. The max point in a square is observed to be the midpoint.

6 References

1. On the product of the distances of a point from the vertices of a polytope. *Israel Journal of Mathematics*, **3**(1), 29–38: 10.1007/BF02760025. Sci. (n.d.-a).

2. The product of the distances of a point inside a regular polytope to its vertices. *Journal of Geometry*, **86**(1-2), 31–41: 10.1007/s00022-006-1813-7. Sci. (n.d.).

7 Appendix

7.1 Research Question 1

7.1.1 Construction of the Fermat Point

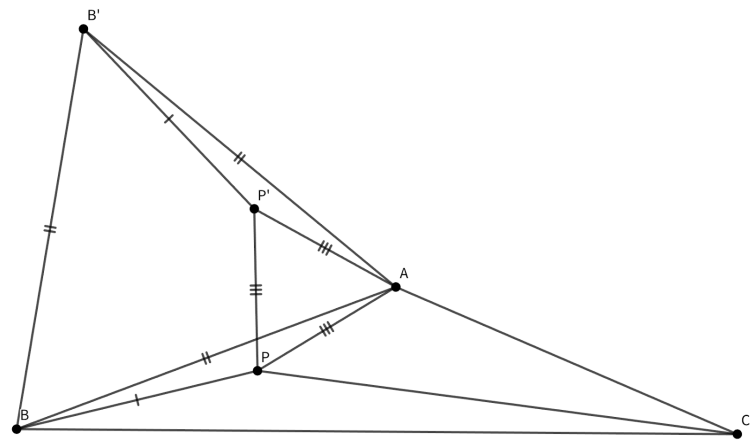


Figure 13: Construction when $\angle BAC \geq 120°$ to find the fermat point in $\triangle ABC$

Case 1: There exists an angle $\geq 120°$ in $\triangle ABC$ Let $|XY|$ denote the distance between points X and Y.
Let point P be any point in triangle $\triangle ABC$.
Rotate $\triangle ABP$ $60°$ clockwise about A. Let the image of point B be B' and point P be P' after rotation, such that B' and C are on opposite sides of the line AB.
We have $\triangle ABP$ is congruent to $\triangle AB'P'$, APP' equilateral, and ABB' equilateral.
Thus $|CP| + |AP| + |BP| = |CP| + |PP'| + |P'B'|$ which is the length of the path $CPP'B'$.
The length of the path $CPP'B'$ is minimised when point $P' = P = A$, i.e. point A, the obtuse angle, is the Fermat point.
The Fermat point cannot be outside the triangle since shifting the point to the closest point on the boundary of the triangle will give a smaller sum of distances.
Return to section 4.1.1 case 1

Case 2: All angles are $< 120°$ in $\triangle ABC$ Similar to Case 1, we rotate $\triangle ABP$ $60°$ around A. Let the image of point B be B' and point P be P' after rotation, such that B' and C lie on opposite sides of the line AB.

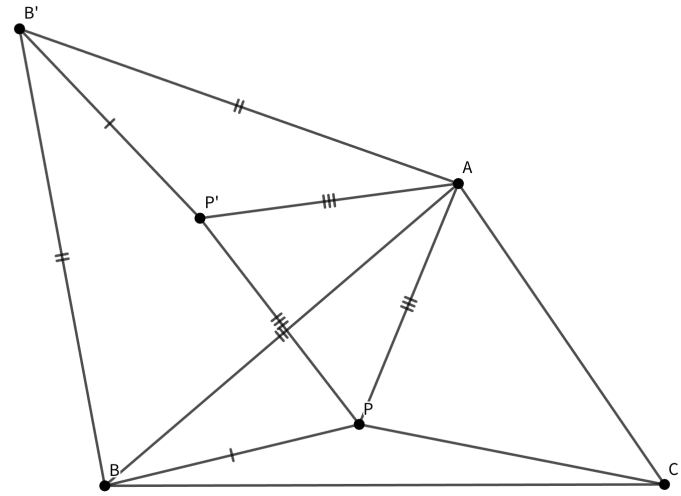


Figure 14: Construction when $\angle BCA, \angle ABC, \angle BAC < 120°$ to find the fermat point in $\triangle ABC$

We have $\triangle ABP$ is congruent to $\triangle AB'P'$, APP' equilateral, and ABB' equilateral.
Thus$|CP| + |AP| + |BP| = |CP| + |PP'| + |P'B'|$ which is the length of the path $CPP'B'$.
The length of the path $CPP'B'$ is minimised when the path is a straight line from C to B'.
Claim: a straight line $CPP'B'$ is obtained when point P = point F, constructed via the following method:
Construct equilateral triangles $\triangle ABB'$ and $\triangle ACC'$ as shown, connect B' to C and C' to B. F is the intersection of BC' and CB'.

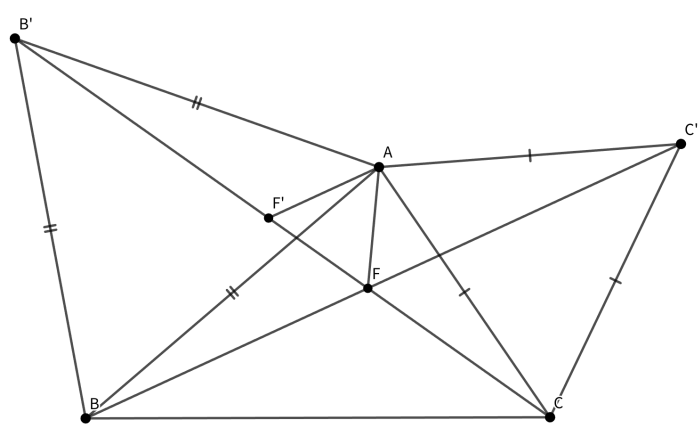


Figure 15

Notice that $\triangle ACB'$ is congruent to $\triangle AC'B$ by rotation.
Thus $\angle ABC' = \angle AB'C$ and $\angle AC'B = \angle ACB'$, hence $AB'BF$ is concyclic and $AC'CF$ is concyclic.
Therefore $\angle AFB' = \angle ABB' = 60°$, hence the rotation of F about A by $60°$ results in F', which lies on the line CB'.

Claim: F is at the intersection of $B'C$, $C'B$, and AD.
Construct the third equilateral triangle $\triangle BCD$. Connect F to D.

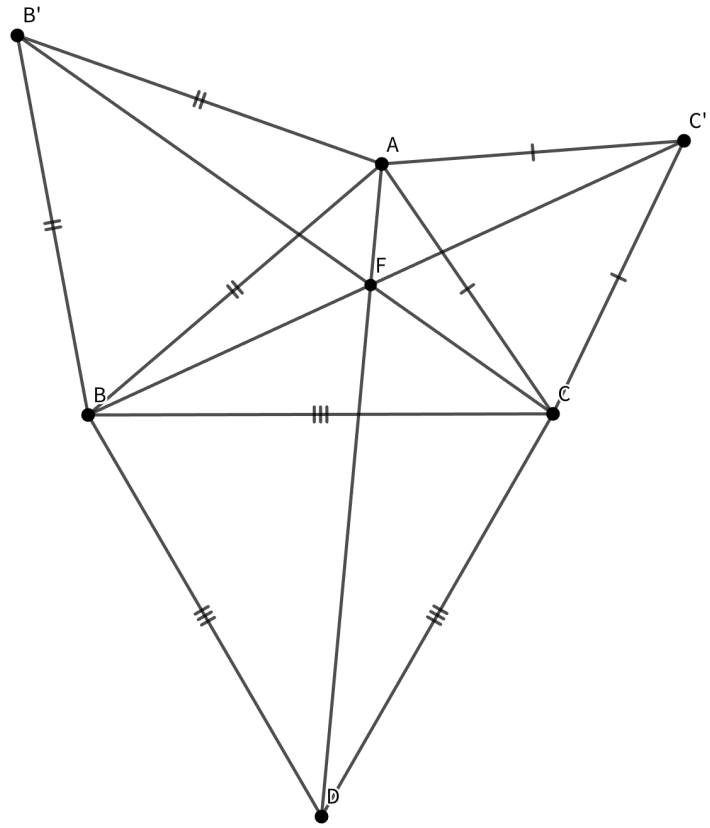


Figure 16: F is a unique point in $\triangle ABC$

$FCDB$ is concyclic, since $\angle BFC = 360° - \angle AFB - \angle CFA = 120°$.
Thus AFD is collinear, since $\angle CFD = 180° - \angle CFA$.
Therefore, F is a unique point in the triangle. Return to section 4.1.1 case 2

7.1.2 General equation for Fermat Point value

Case 1: There exists an angle $\geq 120°$ in $\triangle ABC$ Since point F is at vertex A, Fermat point value $= b + c$

Case 2: All angles are $< 120°$ in $\triangle ABC$ Let

- a, b, and c be the lengths of sides BC, AC, and AB respectively.
- F be the Fermat point
- $\alpha = |AF|$, $\beta = |BF|$, $\gamma = |CF|$ and $S = \alpha + \beta + \gamma$
- $\mathcal{A}$ be the area of triangle ABC

Firstly, as mentioned above, $AB'BF$ is concyclic. Thus, by the inscribed angle theorem, $\angle AFB = 2\angle AB'B = 120°$. Similarly, $\angle AFC = \angle BFC = 120°$.
By the cosine rule,

$$a^2 = \beta^2 + \beta\gamma + \gamma^2$$
$$b^2 = \alpha^2 + \alpha\gamma + \gamma^2$$
$$c^2 = \alpha^2 + \alpha\beta + \beta^2$$

Considering the 3 triangles AFB, AFC, and BFC, the total area of triangle ABC, $\mathcal{A}$, can be calculated as follows:

$$\mathcal{A} = \frac{\sqrt{3}}{4}(\alpha\beta + \beta\gamma + \gamma\alpha)$$

Summing up the first 3 equations and substituting the fourth, we find:

$$a^2 + b^2 + c^2 = 2\left(\alpha^2 + \beta^2 + \gamma^2\right) + \frac{4}{\sqrt{3}}\mathcal{A}$$

$$S^2 = (\alpha + \beta + \gamma)^2 = \left(a^2 + b^2 + c^2 - \frac{4}{\sqrt{3}}\mathcal{A}\right) + \frac{8}{\sqrt{3}}\mathcal{A}$$
$$= \frac{1}{2}\left(a^2 + b^2 + c^2\right) + 2\sqrt{3}\mathcal{A}$$

Finally, using Heron's formula to substitute for the area of triangle ABC in terms of a, b, c, we find Fermat point value $=$

$$\sqrt{\frac{1}{2}(a^2 + b^2 + c^2) + \sqrt{3(a+b+c)(-a+b+c)(a-b+c)(a+b-c)}}$$

Return to section 4.1.2

7.2 Max Point on an Equilateral Triangle

7.2.1 Proof that max point lies on the boundary of the equilateral triangle

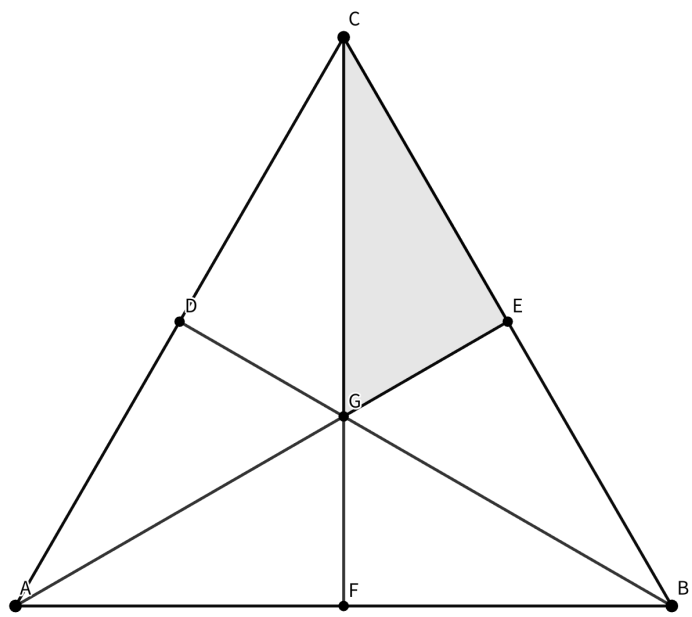


Figure 17: $\triangle ABC$ is equilateral. Consider a variable point P in the triangle. Without loss of generality, P lies in the shaded region, where $G\left(1,\ \frac{1}{\sqrt{3}}\right)$ is the centroid and E is the midpoint of segment CB.

Since P lies in the shaded region, we obtain the bound for h:

$$h \leq \frac{2}{\sqrt{3}}$$

By similar triangles:

$$c \leq |FD| \leq \frac{h}{\sqrt{3}}$$

Claim: On the parallel line ED, the minimum product value is obtained at the midpoint of ED, i.e., when $c = 0$.

Express the distances using Pythagoras's. The product value from vertices to point P is:

$$|AP| \cdot |BP| \cdot |CP| = \sqrt{(1+c)^2 + \left(\sqrt{3}-h\right)^2} \cdot \sqrt{(1-c)^2 + \left(\sqrt{3}-h\right)^2} \cdot \sqrt{h^2+c^2}$$

Differentiate product expression

$$\frac{d}{dc}\left(\sqrt{(1-c)^2 + (\sqrt{3}-h)^2} \cdot \sqrt{(1+c)^2 + (\sqrt{3}-h)^2} \cdot \sqrt{c^2+h^2}\right)$$

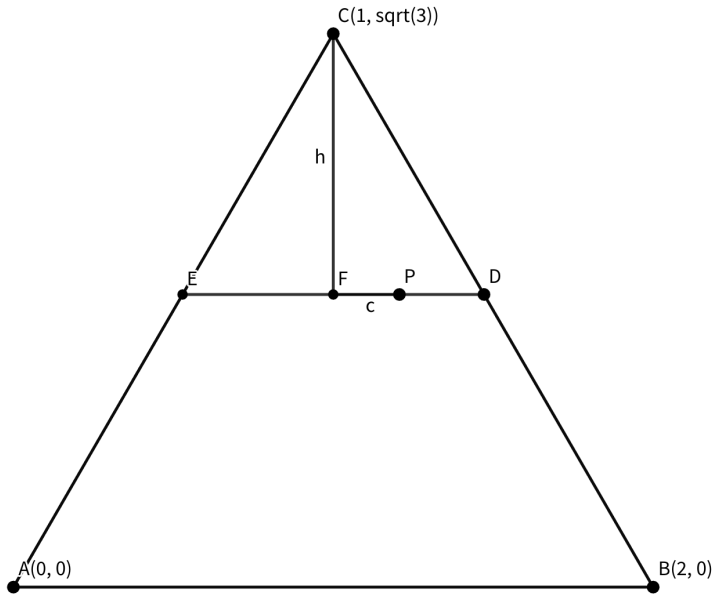


Figure 18: Construct line ED through P such that ED is parallel to AB. Let F be the foot of the perpendicular dropped from C onto ED. Let $h = |CF|$ and $c = |FP|$.

$$
\begin{aligned}
&= \frac{c\sqrt{(1-c)^2 + (\sqrt{3}-h)^2} \cdot \sqrt{(1+c)^2 + (\sqrt{3}-h)^2}}{\sqrt{c^2+h^2}} \\
&+ \frac{(1+c)\sqrt{c^2+h^2} \cdot \sqrt{(1-c)^2 + (\sqrt{3}-h)^2}}{\sqrt{(1+c)^2 + (\sqrt{3}-h)^2}} \\
&+ \frac{(1-c)\sqrt{c^2+h^2} \cdot \sqrt{(1+c)^2 + (\sqrt{3}-h)^2}}{\sqrt{(1-c)^2 + (\sqrt{3}-h)^2}}
\end{aligned}
$$

Solve when derivative equals to zero

$$
\begin{aligned}
&\frac{c\sqrt{(1-c)^2 + (\sqrt{3}-h)^2} \cdot \sqrt{(1+c)^2 + (\sqrt{3}-h)^2}}{\sqrt{c^2+h^2}} \\
&+ \frac{(1+c)\sqrt{c^2+h^2} \cdot \sqrt{(1-c)^2 + (\sqrt{3}-h)^2}}{\sqrt{(1+c)^2 + (\sqrt{3}-h)^2}} \\
&+ \frac{(1-c)\sqrt{c^2+h^2} \cdot \sqrt{(1+c)^2 + (\sqrt{3}-h)^2}}{\sqrt{(1-c)^2 + (\sqrt{3}-h)^2}} = 0
\end{aligned}
$$

$$16c + 8c^3 + 3c^5 - 16\sqrt{3}ch - 8\sqrt{3}c^3h + 24ch^2 + 6c^3h^2 - 8\sqrt{3}ch^3 + 3ch^4 = 0$$

$$c\Big(3c^4 + (8 - 8\sqrt{3}h + 6h^2)c^2 + (16 - 16\sqrt{3}h + 24h^2 - 8\sqrt{3}h^3 + 3h^4)\Big) = 0$$

$$c = 0 \quad \text{or} \quad 3c^4 + (8 - 8\sqrt{3}h + 6h^2)c^2 + (16 - 16\sqrt{3}h + 24h^2 - 8\sqrt{3}h^3 + 3h^4) = 0$$

When

$$3c^4 + \left(8 - 8\sqrt{3}h + 6h^2\right)c^2 + \left(16 - 16\sqrt{3}h + 24h^2 - 8\sqrt{3}h^3 + 3h^4\right) = 0$$

Discriminant Δ is:

$$\begin{aligned}\Delta &= \left(8 - 8\sqrt{3}h + 6h^2\right)^2 - 4(3)\left(16 - 16\sqrt{3}h + 24h^2 - 8\sqrt{3}h^3 + 3h^4\right)\\ &= 64\left(\sqrt{3}h - 2\right)\\ &\Rightarrow \Delta \geq 0\\ &h \geq \frac{2}{\sqrt{3}}\end{aligned}$$

For real solutions, when $h = \frac{2}{\sqrt{3}}$, we also have $c = 0$.
By the bound on h, $c = 0$ is the only real solution.

Take second derivative

$$\begin{aligned}&\frac{d^2}{dc^2}\left(\sqrt{(1-c)^2 + (\sqrt{3}-h)^2} \cdot \sqrt{(1+c)^2 + (\sqrt{3}-h)^2} \cdot \sqrt{c^2+h^2}\right) =\\ &\quad \big(6c^{10} - 9c^8(-4 + 4\sqrt{3}h - 3h^2)\\ &\qquad - 16c^6(-12 + 14\sqrt{3}h - 24h^2 + 8\sqrt{3}h^3 - 3h^4)\\ &\qquad - 6c^4(-32 + 64\sqrt{3}h - 208h^2 + 128\sqrt{3}h^3\\ &\qquad\quad - 140h^4 + 28\sqrt{3}h^5 - 7h^6)\\ &\qquad + 6c^2h^2(64 - 128\sqrt{3}h + 320h^2 - 144\sqrt{3}h^3\\ &\qquad\quad + 112h^4 - 16\sqrt{3}h^5 + 3h^6)\\ &\qquad + h^2(256 - 512\sqrt{3}h + 1472h^2 - 896\sqrt{3}h^3\\ &\qquad\quad + 1120h^4 - 320\sqrt{3}h^5 + 180h^6 - 20\sqrt{3}h^7 + 3h^8)\big)\\ &\quad \Big/ \big((c^2+h^2)^{3/2} \cdot (4 - 2c + c^2 - 2\sqrt{3}h + h^2)^{3/2} \cdot\\ &\qquad (4 + 2c + c^2 - 2\sqrt{3}h + h^2)^{3/2}\big)\end{aligned}$$

When $c = 0$,

$$\frac{d^2}{dc^2} = \frac{1 + (\sqrt{3} - h)^2}{h} + 2h\left(1 - \left(\frac{2}{1 + (\sqrt{3} - h)^2}\right)\right)$$

Claim: Second derivative when $c = 0$ is always non-negative when $h \leq \frac{2}{\sqrt{3}}$
The expression is:

$$\frac{1 + (\sqrt{3} - h)^2}{h} + 2h\left(1 - \left(\frac{2}{1 + (\sqrt{3} - h)^2}\right)\right)$$

Note that $\frac{1+(\sqrt{3}-h)^2}{h}$ is strictly decreasing when h increases from 0 to $\frac{2}{\sqrt{3}}$, since the numerator is strictly decreasing while the denominator is strictly increasing.
Note that $2h\left(1 - \frac{2}{1+(\sqrt{3}-h)^2}\right)$ is strictly decreasing when h increases from $(\sqrt{3} - 1)$ to $\frac{2}{\sqrt{3}}$, as $2h$ is strictly increasing while $\left(1 - \frac{2}{1+(\sqrt{3}-h)^2}\right)$ is negative and strictly decreasing over the same interval.
Hence, the value of the expression is strictly decreasing as h increases from $(\sqrt{3} - 1)$ to $\frac{2}{\sqrt{3}}$.
Note that when $h = \frac{2}{\sqrt{3}}$, the value of the expression is 0.
When $h = (\sqrt{3} - 1)$, the value of the expression is 3.73 (3 s.f.)
Clearly, the expression is greater than 0 when $h < (\sqrt{3} - 1)$ since $\frac{1+(\sqrt{3}-h)^2}{h} > 0$ and $2h\left(1 - \frac{2}{1+(\sqrt{3}-h)^2}\right) > 0$
Thus it is shown that second derivative is always non-negative
Since $c = 0$ is a minimum point and there are no other stationary points, the function is strictly increasing as c increases. Hence the maximum value will be attained at the furthest point from $c = 0$, which is when $c = \frac{h}{\sqrt{3}}$. Return to section 4.2.1

7.2.2 Max point lies on the midpoint of the side of an equilateral triangle

1. Fix midpoint M at $(1, 0)$
2. Say point P shifts by a certain k where $0 < k < 1$
3. Original product at M is $\sqrt{3}$
4. New product is given by the following expression:

$$\begin{aligned}\left(\sqrt{3 + k^2}\right)(1 - k)(1 + k) &= \sqrt{3 + k^2} \cdot (1 - k^2) \\ &= \sqrt{(1 - k^2)^2(3 + k^2)} \\ &= \sqrt{(3 - 2k^2 - k^4)(1 - k^2)}\end{aligned}$$

5. Clearly it is less than $\sqrt{3}$

The maximum product of distances from a point to the 3 vertices in an equilateral triangle ABC is obtained at the midpoint of any side. Return to section 4.2.2

7.3 Research Question 3: Isosceles Triangle

7.3.1 Calculating max point on the base of isosceles triangle

Let $A(0,0), B(2,0), C(1,t)$, i.e. the base length $= 2$ and height of triangle $= t$.

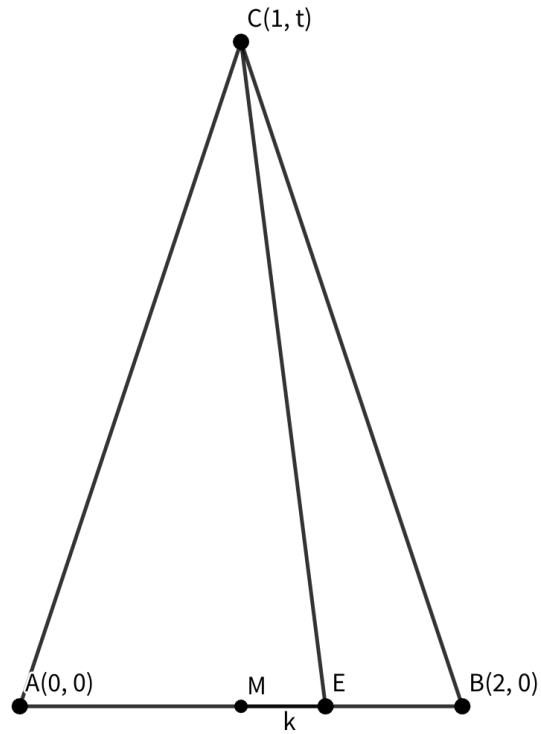


Figure 19: Consider point E on base of isosceles

The product value at E is:

$$|BE| \cdot |AE| \cdot |CE| = \left((1-k)(1+k)\sqrt{k^2+t^2}\right)$$

Differentiate and solve for max

$$\frac{d}{dk}\left((1-k)(1+k)\sqrt{k^2+t^2}\right) = 0$$

$$\frac{(1+k)(1-k)(k)}{\sqrt{k^2+t^2}} + (1-k)\sqrt{k^2+t^2} - (1+k)\sqrt{k^2+t^2} = 0$$

$$(1-k^2)k + (1-k)(k^2+t^2) - (1+k)(k^2+t^2) = 0$$

$$k(-3k^2+1-2t^2) = 0$$

$$\Rightarrow \quad k = 0 \quad \text{or} \quad k = \pm\frac{\sqrt{1-2t^2}}{\sqrt{3}}$$

For there to be real solutions for k when $-3k^2 + 1 - 2t^2 = 0$,

$$t \leq \frac{1}{\sqrt{2}}$$

When $t > \frac{1}{\sqrt{2}}$, $k = 0$ is a maximum point since second derivative when $k = 0$ is

$$\frac{1 - 2t^2}{t} < 0$$

When $t = \frac{1}{\sqrt{2}}$, $k = 0$ is a maximum point, simply by checking values of $k > 0$ and $k < 0$
When $t < \frac{1}{\sqrt{2}}$, $k = 0$ is a minimum point, and $k = \frac{\sqrt{1-2t^2}}{\sqrt{3}}$ is a maximum point as second derivative when

$$k = \frac{\sqrt{1 - 2t^2}}{\sqrt{3}} \quad \text{is} \quad \frac{2\sqrt{3}(-1 + 2t^2)}{\sqrt{1 + t^2}} < 0$$

Substituting $k = 0$ into $(1 - k)(1 + k)\sqrt{k^2 + t^2}$, we obtain max product value $= t$
Substituting $k = \frac{\sqrt{1-2t^2}}{\sqrt{3}}$ into $(1 - k)(1 + k)\sqrt{k^2 + t^2}$, we obtain max product value

$$= \frac{2\sqrt{(1 - 2t^2)^3}}{3\sqrt{3}}$$

7.3.2 Calculating max point on the leg of isosceles triangle

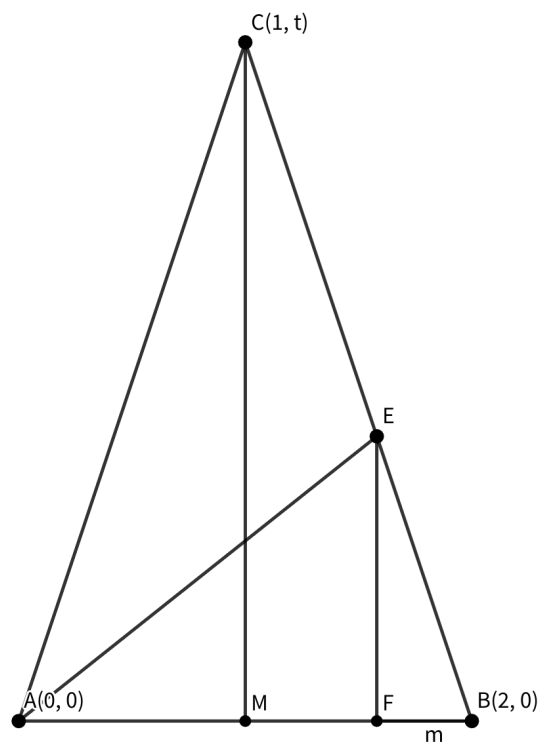


Figure 20: Consider point E on leg of isosceles

$|EF| = mt$ since $\triangle BEF$ is similar to $\triangle BCM$

$$|CB| = \sqrt{t^2+1}$$

Product value at E is $|AE| \cdot |BE| \cdot |CE|$:

$$= \left(\sqrt{(2-k)^2+k^2t^2}\right)\left(k\sqrt{1+t^2}\right)\left(\sqrt{(1-k)^2+(t-kt)^2}\right)$$

$$= k\sqrt{1+t^2}\sqrt{(2-k)^2+k^2t^2}\sqrt{(1-k)^2+(t-kt)^2}$$

Maximum product value is achieved when derivative of the expression = 0

$$\frac{d}{dk}\left(k\sqrt{1+t^2}\sqrt{(2-k)^2+k^2t^2}\sqrt{(1-k)^2+(t-kt)^2}\right) = 0$$

$$\frac{k\sqrt{1+t^2}\left((2-k)^2+k^2t^2\right)\left(-2(1-k)-2t(1-kt)\right)}{2\sqrt{(1-k)^2+(t-kt)^2}}$$

$$+\frac{k\sqrt{1+t^2}\left(-2(2-k)+2kt^2\right)\sqrt{(1-k)^2+(t-kt)^2}}{2\sqrt{(2-k)^2+k^2t^2}}$$

$$+\sqrt{1+t^2}\sqrt{(2-k)^2+k^2t^2}\sqrt{(1-k)^2+(t-kt)^2} = 0$$

$$3k^3(1+t^2) - 2k^2(6+t^2) + 14k - 4 = 0$$

Make k the subject:
Since $0 < k < 1$,

$$k = \frac{2(t^2+6)}{9(1+t^2)} + \frac{\sqrt[3]{3\sqrt{2}\left(4t^6+126t^4-405t^2+9\sqrt{3}\sqrt{8t^{10}-24t^8+102t^6+305t^4+168t^2-3-27}\right)}}{9(1+t^2)} - \frac{(-4t^4+78t^2-18)}{9(1+t^2)\sqrt[3]{2(1+t^2)}\sqrt[3]{4t^6+126t^4-405t^2+9\sqrt{3}\sqrt{8t^{10}-24t^8+102t^6+305t^4+168t^2-3-27}}} \tag{1}$$

Thus, for any positive value t, there is only one real value of k, and hence one stationary point, and from checking the graph it is a maximum point.

7.3.3 Graph analysis

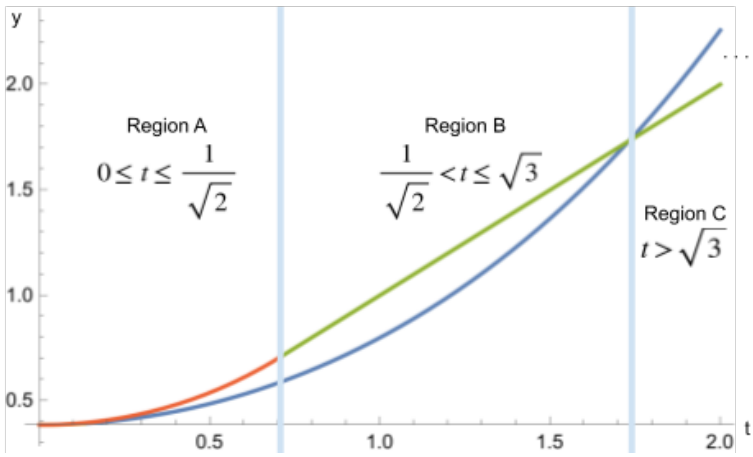


Figure 21: Red line and Green line represent the maximum product of distances to vertices when point E is on base. Blue line represents max product of distances to vertices when point F is on side.

Equation of red line is

$$y = \frac{2\sqrt{(1+2t^2)^3}}{3\sqrt{3}}, \quad \text{with maximum product value on the base when } t < \frac{1}{\sqrt{2}}$$

Equation of green line is

$$y = t, \quad \text{with maximum product value on base when } t > \frac{1}{\sqrt{2}}$$

Equation of blue line is

$$y = (1-k)k(1+t^2)\sqrt{(1+k)^2 + (1-k)^2 t^2}, \quad \text{maximum of which occurs when}$$

Since $0 < k < 1$, k takes on the value (1)

$$-\frac{-4t^4 + 78t^2 - 18}{9\sqrt[3]{2(t^2+1)}\sqrt[3]{4t^6 + 126t^4 - 405t^2 + 9\sqrt{3}\sqrt{8t^{10} - 24t^8 + 102t^6 + 305t^4 + 168t^2 - 3} - 27}}$$

7.4 Research Question 4: General Triangle

7.4.1 Calculating the max point on each side

Let the three vertices be $A(0,0), B(2,0), C(x,y)$

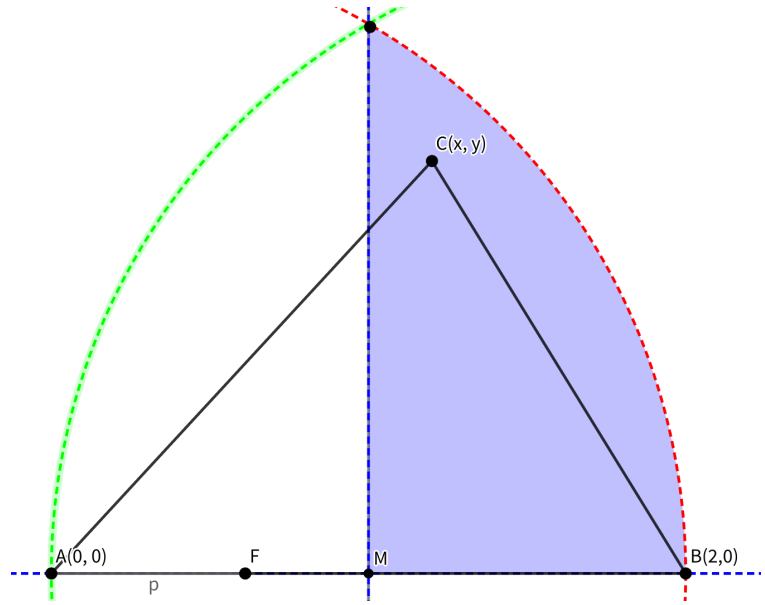


Figure 22: Without loss of generality, $|BA| \geq |AC| \geq |CB|$.

C lies in the shaded region represented by:

$$x^2 + y^2 \leq 4 \quad \text{(red circle)}$$
$$(x-2)^2 + y^2 \leq 4 \quad \text{(green circle)}$$
$$1 \leq x < 2, \quad y > 0$$

—

Claim: with the above set-up, the max point F along AB lies on the left of the midpoint M.

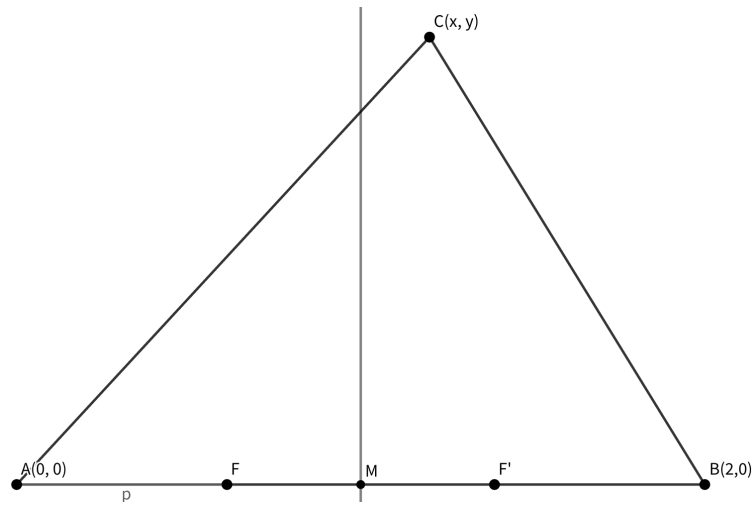


Figure 23: It is shown that the product value at $F' < F$

Let p be the distance $|BF'|$ Suppose that F' is the max point along AB that lies on the right of midpoint M. Reflect F' across the perpendicular bisector to obtain point F. Then $|BF'| = |AF| = p$
Product at point F is always greater than product at point F' since:

$$\begin{aligned}
&|AF| \cdot |BF| \cdot |CF| > |AF'| \cdot |BF'| \cdot |CF'| \\
\Leftrightarrow\ &(p)(2-p)|CF| > (2-p)(p)|CF'| \\
\Leftrightarrow\ &|CF| > |CF'| \\
\Leftrightarrow\ &\sqrt{(x-p)^2+y^2} > \sqrt{(2-x-p)^2+y^2} \\
\Leftrightarrow\ &(x-p)^2 > (2-x-p)^2 \\
\Leftrightarrow\ &x+p > 1 \quad \text{which is trivially true by setup.}
\end{aligned}$$

Product value at $F = p(2-p)\sqrt{(x-p)^2+y^2}$
Maximum product value is at F when the derivative of the expression $= 0$, since a lower product value will be achieved at A and F'.

$$\begin{aligned}
&\frac{d}{dp}\left(p(2-p)\sqrt{(x-p)^2+y^2}\right) = 0 \\
\Rightarrow\ &-\frac{(2-p)p(-p+x)}{\sqrt{(-p+x)^2+y^2}} + (2-p)\sqrt{(-p+x)^2+y^2} - p\sqrt{(-p+x)^2+y^2} = 0 \\
\Rightarrow\ &-3p^3 + p^2(4+5x) - 2p(3x+x^2+y^2) + 2(x^2+y^2) = 0 \qquad (2)
\end{aligned}$$

Due to the bounds on the variable p, as well as bounds on the constants x, y, we obtain only one real root for p, which is also supported by graphing the product value against p — showing the curve only has one maximum point.
The exact form for p is too long to include, it is simply the root of the cubic equation (2), where x and y are the coordinates of the variable vertice C, thus we will represent the root of the cubic equation as p'.

Substituting $x = 1$, i.e., the isosceles triangle case, yields the same result as derived in section 4.3.1, hence this result is a generalisation of section 4.3.1.
Similarly, for sides CA and BC, we set up the product expression, differentiate, solve derivative $= 0$, and substitute the max point back into the product expression to obtain their maximums. Return to section 4.4.1

7.4.2 Comparing maximums on each side of general triangle

A 3D graph is used to compare the maximum values of the product of distances when the point lies on AB, BC and CA.

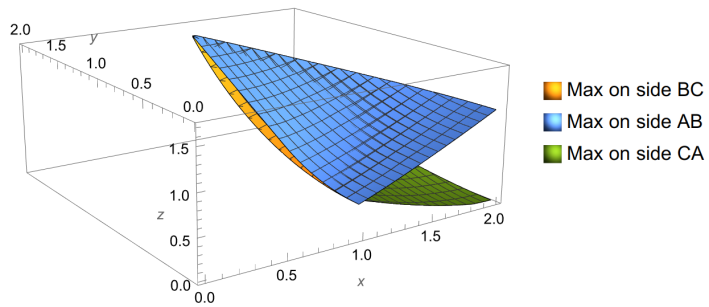


Figure 24: 3dgraph?

Note that the z-axis represents the product value, while the x-y plane represents the coordinates of points. (Refer to table below)

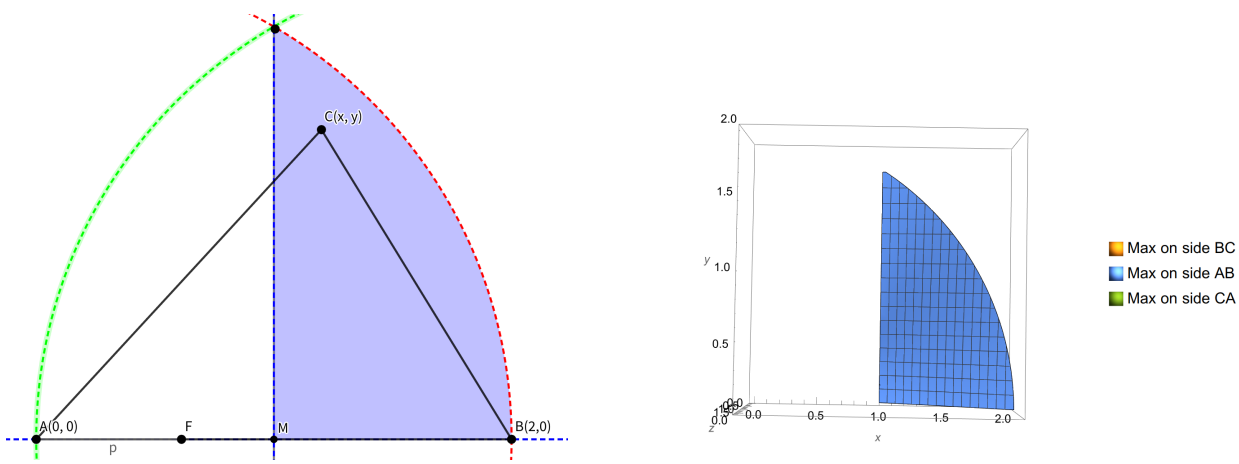


Figure 25: The x-y coordinates in the 3D graph correspond with the 2D graph on the left.

Note that only the shaded area in the figure above needs to be considered as the triangle can be simply reflected along the line $x = 1$
The planes intersect at $(x, y, z) = (1, \sqrt{3}, \sqrt{3})$This corresponds with the equilateral case, solved in section 4.2.

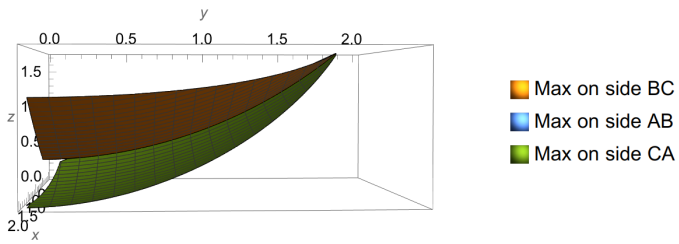


Figure 26: sideviewgeneral?

When $x \to 2$ and $y \to 0$, the green plane approaches 0,while the orange and blue planes get infinitely close, corresponding to the degenerate triangle case where point $C = B$.
It can be observed that the blue plane which represents the max on side AB, the longest side, is always on top, i.e. max point lies on the longest side.
Hence, the solution for p on the longest side, found as the real root of (2), is the max point in a general triangle.
Return to section 4.4.2